Artificial Intelligence

Probabilistic Inference

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Approximate Inference for Bayesian Networks

Exact inference in large Bayesian networks is intractable, so we need approximate inference methods. The methods we'll learn are randomized sampling agorithms, a.k.a., **Monte Carlo** algorithms, a.k.a., **stochastic simulation**. They work by

- generating random events based on the probabilities in the Bayes net and
- counting up the different answers found in those random events.

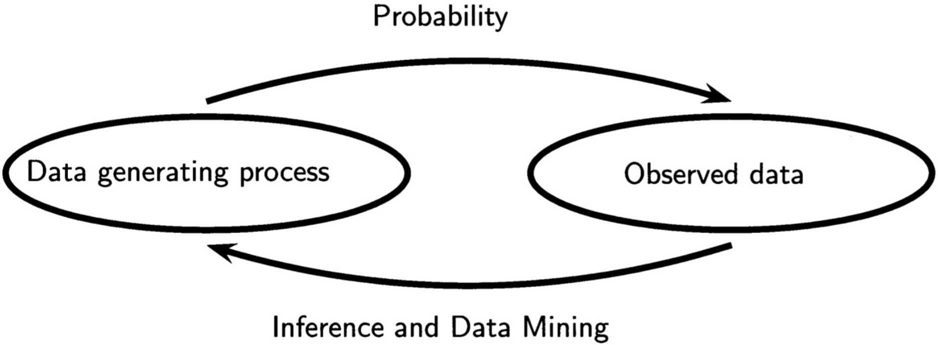
The more the samples, the closer to the true probability distribution.

We'll learn two families of Monte carlo algorithms:

- direct sampling, and
- Markov chain sampling.

The Utility of Sampling

We observe data generated by some process.



[1]

Then we use those samples to make inferences about the data generating process.

- In supervised machine learning we assume a distribution for the data generating process and estimate its parameters.
- In probabilistic inference, we have a model of the full joint distribution (a Bayes net) and use it to generate samples to answer probabilistic queries.

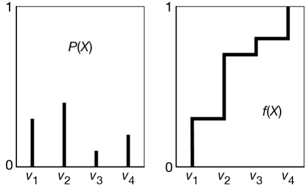
[1] All of Statistics

Computational Sampling of a Univariate Distribution

The primitive element in any sampling algorithm is the generation of samples from a known probability distribution.

To generate samples from a single discrete or real-valued variable, X, first totally order the values in the domain of X.

- For discrete variables, if there is no natural order, just create an arbitrary ordering. Given this ordering, the cumulative probability distribution is a function of x defined by $f(x) = P(X \leq x)$.



To generate a random sample for X:

- Select a uniform random number y in the domain $[0, 1]$.
- Let v be the value of X that maps to y in the cumulative probability distribution. That is, v is the element of $domain(X)$ such that
  - $f(v) = y$ or, equivalently, $v = f^{-1}(y)$

Then, $X = v$ is a random sample of X, chosen according to the distribution of X.

This procedure works for a single variable. If we have multiple variables and a Bayes net to represent the joint distribution over them, there are several algorithms

Prior Sampling, a.k.a., Forward Sampling

To sample from a Bayes net with no associated evidence, sample each node in topological order.

- Sample unconditioned nodes according to their prior distributions.
  - Samples become fixed values for sampling CPTs of child nodes.
- Continue sampling child nodes, in order, until all variables are sampled.

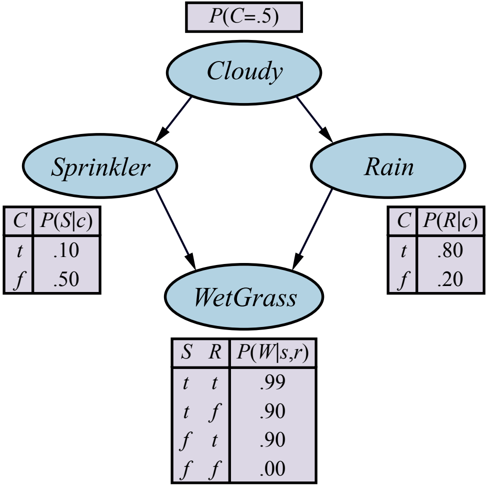
function PRIOR-SAMPLE(*bn*) **returns** an event sampled from the prior specified by *bn*
  inputs: *bn*, a Bayesian network specifying joint distribution $\mathbf{P}(X_1, \ldots, X_n)$

  $\mathbf{x} \leftarrow$ an event with n elements
  for each variable X_i **in** $X_1, \ldots, X_n$ **do**
    $\mathbf{x}[i] \leftarrow$ a random sample from $\mathbf{P}(X_i \mid parents(X_i))$
  return x

Example: Generating a Sample from Sprinkler Bayes Net

1. Sample from $P(Cloudy) = \langle 0.5, 0.5 \rangle$
   - Get value of true.
2. Sample from $P(Sprinkler \mid c) = \langle 0.1, 0.9 \rangle$
   - Get value of false.
3. Sample from $P(Rain \mid c) = \langle 0.8, 0.2 \rangle$
   - Get value of true.
4. Sample from $P(WetGrass \mid \neg s, r) = \langle 0.9, 0.1 \rangle$
   - Get value of true.

Full sampled event: $[true, false, true, true]$



From Sampling to Probability Estimates

A Bayes net represents a full joint distribution, so a sample generated from a Bayes net is a sample from the prior joint distribution over all the variables. So, if S_{PS} is a sample generated by the PRIOR-SAMPLE algorithm, then:

$$S_{PS}(x_1, \ldots, x_n) = \prod_{i=1}^{n} Pr\left(x_i \mid parents(X_i)\right) = P(x_1, \ldots, x_n)$$

To estimate these probabilities using samples, we simply generate N samples, count the number of events of interest among the N samples, and divide that number by N. This should converge to the true probability:

$$\lim_{N \to \infty} \frac{N_{PS}(x_1, \ldots, x_n)}{N} = S_{PS}(x_1, \ldots, x_n) = P(x_1, \ldots, x_n)$$

From the CPTs in the Bayes net, the probability of the example event we sampled, $[true, false, true, true]$ is:

$$S_{PS}(true, false, true, true) = 0.5 \cdot 0.9 \cdot 0.8 \cdot 0.9 = 0.324.$$

So if we generated $N = 1000$ samples, we would expect to see $[true, false, true, true]$ 324 times.

Consistent Probability Estimates

An estimate that becomes exact in the large-sample limit is called **consistent**. A consistent estimate of the probability of any partially speciified event $x_1, \ldots, x_m$ for $x \leq n$ is:

$$P(x_1, \ldots, x_m) \approx \frac{N_{PS}(x_1, \ldots, x_m)}{N} = \hat{P}(x_1, \ldots, x_m) \tag{13.6}$$

That is, $\hat{P}(x_1, \ldots, x_m)$ is an approximation of $P(x_1, \ldots, x_m)$ that converges to the true probability as N approaches ∞

- Forward sampling generates *prior* probabilities, that is, probabilities of events prior to observing evidence.
- What we usually want is *posterior* probabilities, $P(X \mid \boldsymbol{e})$ – probability of X given that we have observed evidence $\boldsymbol{e}$.

We now turn to algorithms for computing posteriors, which use prior sampling as a subroutine.

Rejection Sampling

Rejection sampling is a general method for producing samples from a hard-to-sample distribution given an easy-to-sample distribution. In its simplest form, it can be used to compute conditional probabilities, that is, to determine $P(X \mid \boldsymbol{e})$.

Let

- $\hat{\boldsymbol{P}}(X \mid \boldsymbol{e})$ be a vector of the probabilities for each value of $x \in X$ and
- $\boldsymbol{N}_{PS}(X, \boldsymbol{e})$ be a vector of sample counts for each $x \in X$ where the sample is consistent with the evidence $\boldsymbol{e}$ (the variable **C** in the algorithm).

Then:

$$\hat{\boldsymbol{P}}(X \mid \boldsymbol{e}) = \alpha \boldsymbol{N}_{PS}(X, \boldsymbol{e}) = \frac{\boldsymbol{N}_{PS}(X, \boldsymbol{e})}{\boldsymbol{N}_{PS}(\boldsymbol{e})}$$

function REJECTION-SAMPLING(X, **e**, bn, N) **returns** an estimate of $\mathbf{P}(X \mid \mathbf{e})$
  inputs: X, the query variable
        e, observed values for variables **E**
        bn, a Bayesian network
        N, the total number of samples to be generated
  local variables: **C**, a vector of counts for each value of X, initially zero

  for j = 1 **to** N **do**
    x ← PRIOR-SAMPLE(bn)
    if x is consistent with **e then**
      C[j] ← **C**[j]+1 where x_j is the value of X in **x**
  return NORMALIZE(**C**)

Rejection Sampling Example

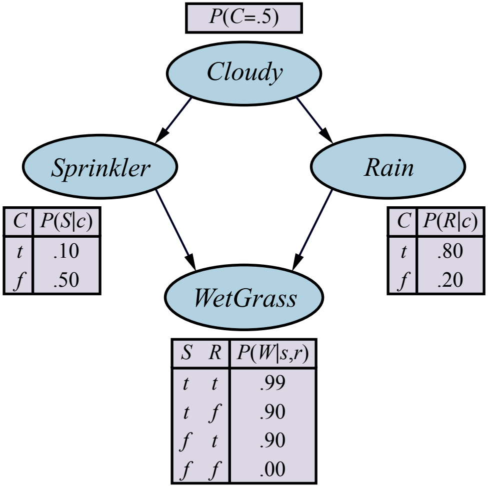
Say we use prior sampling to generate 100 samples and we want to estimate $\boldsymbol{P}(Rain \mid Sprinkler = true)$.

- 73 samples have $Sprinkler = false$, so are *rejected*.
- Of the 27 *accepted* samples:
  - 8 samples have $Rain = true$
  - 19 samples have $Rain = false$.

Then:

$$\hat{\boldsymbol{P}}(Rain \mid Sprinkler = true) = \frac{\boldsymbol{N}_{PS}(X, \boldsymbol{e})}{\boldsymbol{N}_{PS}(\boldsymbol{e})} = \frac{\langle 8, 19 \rangle}{27} = \langle 0.296, 0.704 \rangle$$

Calculating directly from the CPTs in the Bayes net would give us $\langle 0.3, 0.7 \rangle$.



Importance Sampling

The general statistical technique of **importance sampling** aims to emulate the effect of sampling from a distribution P using samples from another distribution Q whose samples are easier to obtain.

We ensure that the answers are correct in the limit by applying a correction factor $\frac{P(\boldsymbol{x})}{Q(\boldsymbol{x})}$, also known as a **weight**, to each sample $\boldsymbol{x}$ when counting up the samples.

Our query variable will always be one of the nonevidence variables, Z. The idea of importance sampling is to sample from $P(\boldsymbol{z} \mid \boldsymbol{e})$:

$$\hat{P}(\boldsymbol{z} \mid \boldsymbol{e}) = \frac{N_P(\boldsymbol{z})}{N} \approx P(\boldsymbol{z} \mid \boldsymbol{e})$$

but using an arbitrary distribution $Q(\boldsymbol{z})$:

$$\hat{P}(\boldsymbol{z} \mid \boldsymbol{e}) = \frac{N_Q(\boldsymbol{z})}{N} \frac{P(\boldsymbol{z} \mid \boldsymbol{e})}{Q(\boldsymbol{z})} \approx Q(\boldsymbol{z}) \frac{P(\boldsymbol{z} \mid \boldsymbol{e})}{Q(\boldsymbol{z})} = P(\boldsymbol{z} \mid \boldsymbol{e})$$

The question is: which Q to use? We want Q as close as possible to $P(\boldsymbol{z} \mid \boldsymbol{e})$ with $Q(\boldsymbol{z}) \neq 0$ whenever $P(\boldsymbol{z} \mid \boldsymbol{e}) \neq 0$. The most common approach is **likelihood weighting** . . .

Likelihood Weighting

Fix the values for the evidence variables $\boldsymbol{E}$ and sample all the nonevidence variables in topological order, each conditioned on its parents – guarantees each sample consistent with evidence.

Let the sampling distribution be Q_{WS} and nonevidence variables be $\boldsymbol{Z} = \{Z_1, \ldots, Z_l\}$. Then:

$$Q_{WS}(\boldsymbol{z}) = \prod_{i=1}^{l} P(z_i \mid parents(Z_i))$$

Now we need to know the weight for each sample. By general scheme for importance sampling:

$$w(\boldsymbol{z}) = \frac{P(\boldsymbol{z} \mid \boldsymbol{e})}{Q_{WS}(\boldsymbol{z})} = \alpha \frac{P(\boldsymbol{z}, \boldsymbol{e})}{Q_{WS}(\boldsymbol{z})}$$

where the normalizing factor $\alpha = \frac{1}{P(e)}$ is the same for all samples. Since $\boldsymbol{z}$ and $\boldsymbol{e}$ cover all the variables in the Bayes net, $P(\boldsymbol{z}, \boldsymbol{e})$ is the product of the conditional probabilities for the nonevidence variables times the product of the conditional probabilities for the evidence variables:

$$\begin{aligned} w(\boldsymbol{z}) = \alpha \frac{P(\boldsymbol{z}, \boldsymbol{e})}{Q_{WS}(\boldsymbol{z})} &= \alpha \frac{\prod_{i=1}^{l} P(z_i \mid parents(Z_i)) \prod_{i=1}^{m} P(e_i \mid parents(E_i))}{\prod_{i=1}^{l} P(z_i \mid parents(Z_i))} \\ &= \alpha \prod_{i=1}^{m} P(e_i \mid parents(E_i)) \qquad (14.9) \end{aligned}$$

The name of this method comes from the fact that probabilities of evidence are generally called likelihoods.

Likelihood Weighting Algorithm

function LIKELIHOOD-WEIGHTING($X, \mathbf{e}, bn, N$) **returns** an estimate of $\mathbf{P}(X \mid \mathbf{e})$
  inputs: X, the query variable
    $\mathbf{e}$, observed values for variables $\mathbf{E}$
    bn, a Bayesian network specifying joint distribution $\mathbf{P}(X_1, \ldots, X_n)$
    N, the total number of samples to be generated
  local variables: $\mathbf{W}$, a vector of weighted counts for each value of X, initially zero

  for j = 1 **to** N **do**
    $\mathbf{x}, w \leftarrow$ WEIGHTED-SAMPLE($bn, \mathbf{e}$)
    $\mathbf{W}[j] \leftarrow \mathbf{W}[j] + w$ where x_j is the value of X in $\mathbf{x}$
  return NORMALIZE($\mathbf{W}$)

function WEIGHTED-SAMPLE($bn, \mathbf{e}$) **returns** an event and a weight

  $w \leftarrow 1$; $\mathbf{x} \leftarrow$ an event with n elements, with values fixed from $\mathbf{e}$
  for i = 1 **to** n **do**
    if X_i is an evidence variable with value x_{ij} in $\mathbf{e}$
      then $w \leftarrow w \times P(X_i = x_{ij} \mid parents(X_i))$
      else $\mathbf{x}[i] \leftarrow$ a random sample from $\mathbf{P}(X_i \mid parents(X_i))$
  return $\mathbf{x}$, w

Likelihood Weighting Example

Given the query

- $P(Rain \mid Cloudy = true, WetGrass = true)$

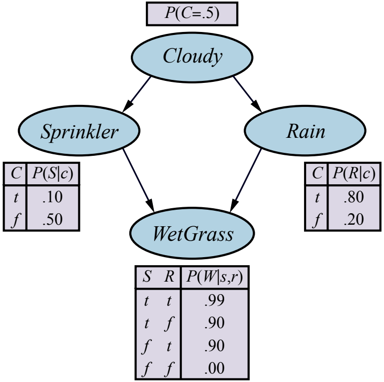
and the ordering

- $Cloudy, Sprinkler, Rain, WetGrass,$

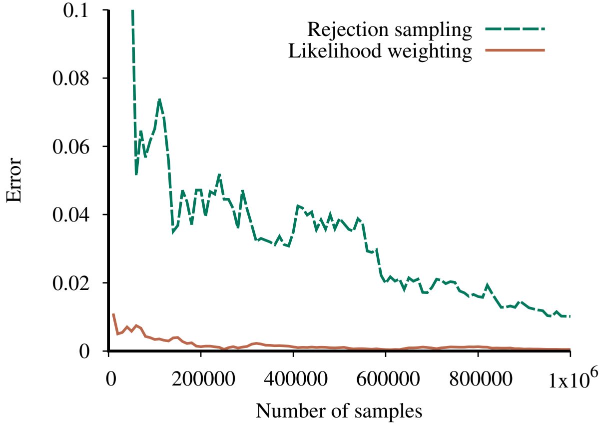
we initialize $w = 1.0$ and proceed as follows:

1. $Cloudy$ is an evidence variable with value true. Therefore, we set
$$w \leftarrow w \cdot P(c) = 0.5.$$
2. $Sprinkler$ is not an evidence variable, so sample from $P(Sprinkler \mid Cloudy = true) = \langle 0.1, 0.9 \rangle$; suppose this returns false.
3. $Rain$ is not an evidence variable, so sample from $P(Rain \mid Cloudy = true) = \langle 0.8, 0.2 \rangle$; suppose this returns true.
4. $WetGrass$ is an evidence variable with value true. Therefore, we set
$$w \leftarrow w \times P(WetGrass = true \mid \neg s, r) = 0.5 \cdot 0.9 = 0.45.$$

Here WEIGHTED-SAMPLE returns the event $[true, false, true, true]$ with weight 0.45, which we tally under $Rain = true$.



Rejection vs. Importance Sampling



Markov Chain Monte Carlo (MCMC) Algorithms

Instead of generating each sample from scratch, MCMC algorithms generate a sample by making a random change to the preceding sample.

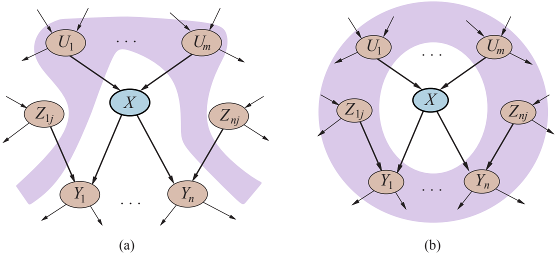
Think of an MCMC algorithm as being in a particular current state that specifies a value for every variable and generating a next state by making random changes to the current state.

We'll learn two MCMC algorithms:

- Gibbs sampling, and
- Metropolis-Hastings

Gibbs Sampling and Markov Blankets

Conditional Independence Relations

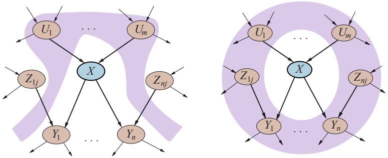


- (a) Each variable is conditionally independent of its non-descendants, given its parents.
- (b) A variable is conditionally independent of all other nodes in the network, given its parents, children, and children's parents – that is, given its **Markov blanket**.

For example, the variable $Burglary$ is independent of $JohnCalls$ and $MaryCalls$, given $Alarm$ and $Earthquake$.

Gibbs Sampling

- Fix the evidence variables.
- Start in a arbitrary state sampled from the nonevidence variables.
  - Each X_i is sampled from the values in its Markov blanket, $mb(X_i)$



(a) (b)

$$P(x_i \mid mb(X_i)) = \alpha P(x_i \mid parents(X_i)) \prod_{Y_j \in Children(X_i)} P(y_i \mid parents(Y_j)) \tag{13.10}$$

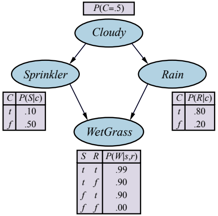
- Wander the state space randomly, keeping the evidence variables fixed and flipping one nonevidence variable by sampling from a distribution calculated Equation 13.10.

Gibbs sampling for X_i means sampling conditioned on the current values of the variables in its Markov blanket.

Gibbs Sampling Example: Step 1

Consider the query $P(Rain \mid Sprinkler = true, WetGrass = true)$.

- Evidence variables $Sprinkler$ and $WetGrass$ fixed to observed values (true), nonevidence variables $Cloudy$ and $Rain$ initialized randomly to, say, true and false respectively.
  - Initial state is $[true, \mathbf{true}, false, \mathbf{true}]$. Fixed evidence variables marked in bold.



Now the nonevidence variables Z_i are sampled repeatedly in some random order according to a probability distribution $\rho(i)$ for choosing variables. For example:

1. $Cloudy$ is chosen and then sampled, given the current values of its Markov blanket: in this case, we sample from $P(Cloudy|Sprinkler = true, Rain = false)$ whose distribution is calculated using Equation 13.10:

$$P(x_i \mid mb(X_i)) = \alpha P(x_i \mid parents(X_i)) \prod_{Y_j \in Children(X_i)} P(y_i \mid parents(Y_j)) \qquad (13.10)$$

$$P(c \mid s, \neg r) = \alpha P(c) P(s \mid c) P(\neg r \mid c) = \alpha 0.5 \cdot 0.1 \cdot 0.2$$

$$P(\neg c \mid s, \neg r) = \alpha P(\neg c) P(s \mid \neg c) P(\neg r \mid \neg c) = \alpha 0.5 \cdot 0.5 \cdot 0.8$$

yielding $\alpha\langle 0.001, 0.020\rangle \approx \langle 0.048, 0.952\rangle$

- Suppose the result is $Cloudy = false$. Then the new current state is $[false, \mathbf{true}, false, \mathbf{true}]$.

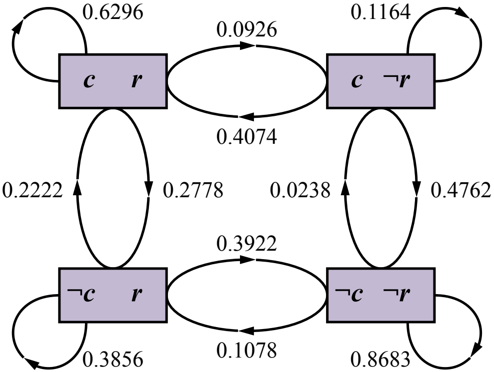
Gibbs Sampling Example: Step 2

2. Sampling $\rho(i)$ then chooses $Rain$ as next variable. We sample from $P(Rain \mid Cloudy = false, Sprinkler = true, WetGrass = true)$ using Equation 13.10 (just like Step 1, not shown here).

- Suppose this yields $Rain = true$. The new current state is $[false, \textbf{true}, true, \textbf{true}]$.

Figure on the right shows the complete Markov chain for the case where variables are chosen uniformly, i.e., $\rho(Cloudy) = \rho(Rain) = 0.5$.

- The algorithm wanders around in this graph, following links with stated probabilities.
- Each state visited during this process is a sample that contributes to the estimate for the query variable Rain.
  - If the process visits 20 states where Rain is true and 60 states where Rain is false, then the answer to the query is $NORMALIZE(\langle 20, 60 \rangle) = \langle 0.25, 0.75 \rangle$.



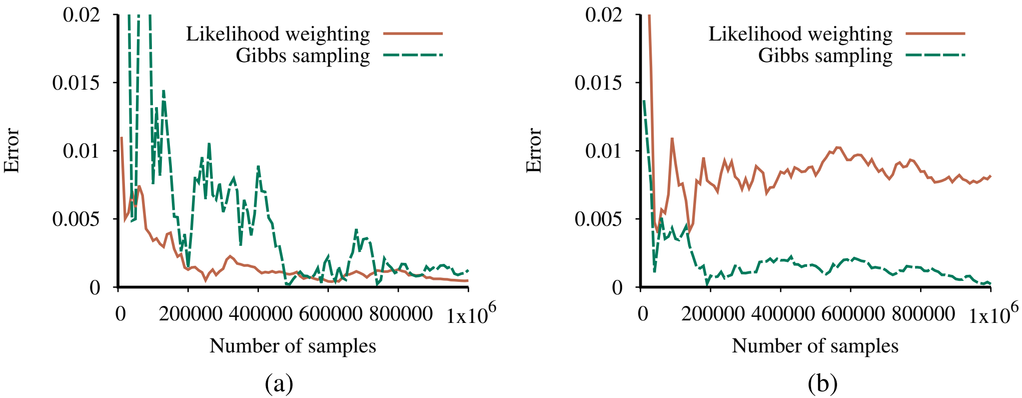
Gibbs Sampling Algorithm

function GIBBS-ASK($X, \mathbf{e}, bn, N$) **returns** an estimate of $\mathbf{P}(X \mid \mathbf{e})$
local variables: $\mathbf{C}$, a vector of counts for each value of X, initially zero
$\mathbf{Z}$, the nonevidence variables in bn
$\mathbf{x}$, the current state of the network, initialized from $\mathbf{e}$

initialize $\mathbf{x}$ with random values for the variables in $\mathbf{Z}$
for $k = 1$ **to** N **do**
choose any variable Z_i from $\mathbf{Z}$ according to any distribution $\rho(i)$
set the value of Z_i in $\mathbf{x}$ by sampling from $\mathbf{P}(Z_i \mid mb(Z_i))$
$\mathbf{C}[j] \leftarrow \mathbf{C}[j] + 1$ where x_j is the value of X in $\mathbf{x}$
return NORMALIZE($\mathbf{C}$)

Gibbs Sampling vs. Importance Sampling

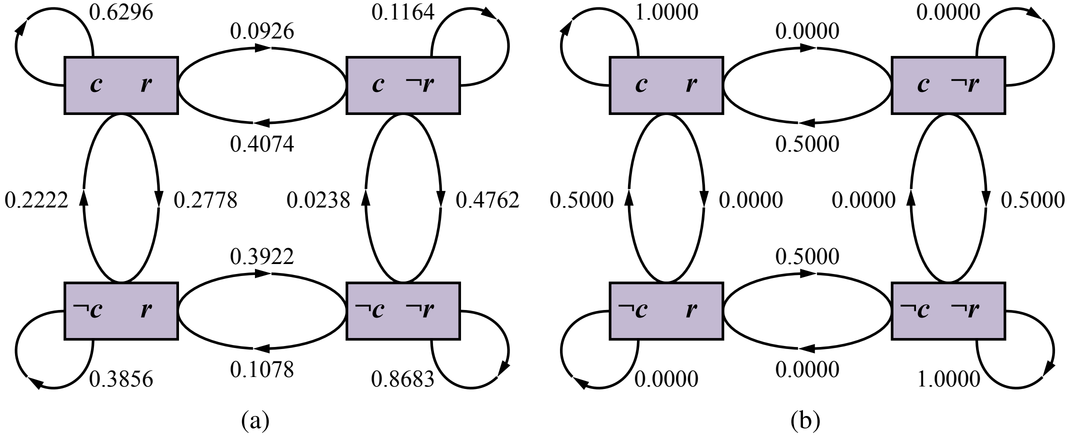
Performance of Gibbs sampling compared to likelihood weighting on the car insurance network.



- (a) for the standard query on $PropertyCost$, and
- (b) for the case where the output variables are observed and Age is the query variable.

Gibbs sampling typically outperforms likelihood weighting when evidence is mostly downstream.

Markov Chains



- (a) The states and transition probabilities of the Markov chain for the query $P(Rain|Sprinkler = true, WetGrass = true)$. Note the self-loops: the state stays the same when either variable is chosen and then resamples the same value it already has.
- (b) The transition probabilities when the CPT for $Rain$ constrains it to have the same value as $Cloudy$.

Metropolis


METROPOLIS
MANUSKRIPT:
THEA VON HARBOU
MUSIK:
GOTTFRIED HUPPERTZ
EIN FILM VON FRITZ LANG
JN DEN HAUPTROLLEN:
BRIGITTE HELM · GUSTAV FRÖHLICH.
ALFRED ABEL, RUDOLF KLEIN-ROGGE, THEODOR LOOS, FRITZ RASP, HEINRICH GEORGE
AN DER KAMERA: KARL FREUND, GÜNTHER RITTAU
UFA FILM IM VERLEIH DER PARUFAMET

Metropolis-Hastings Sampling

Like simulated annealing, MH has two stages in each iteration of the sampling process:

1. Sample a new state x' from a **proposal distribution** $q(x'|x)$, given the current state x.
2. Probabilistically accept or reject x' according to the **acceptance probability**

$$a(x'|x) = \min\left(1, \frac{\pi(x')q(x|x')}{\pi(x)q(x'|x)}\right)$$

If the proposal is rejected, the state remains at x.

$q(x'|x)$ could be defined as follows:

- With probability 0.95, perform a Gibbs sampling step to generate x'.
- Otherwise, generate x' by running WEIGHTED-SAMPLE using likelihood weighting.

This has the effect of getting MH "unstuck."